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MILITARY SPECIFICATION

FILTER, BANDPASS (NAVORD DWG 1563728)

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of a Bandpass Filter, NAVORD Dwg 1563728, which is used in the enabler assembly of an acoustic panel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621	Box, Wood, Nailed and Lock-corner

Military

MIL-R-94	Resistors, Variable Composition, General Specification for
MIL-P-116	Preservation, Methods of
MIL-R-10509	Resistors, Fixed Film (High Stability), General Specification for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

FSC 1355

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40″ x 48″ pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Naval Ordance Systems Command

NAVORD Dwg 1563728	Filter, Bandpass (Specification Control Drawing)

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Threads Standsrds for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

United States of America Standards Institute

USAS Y32.2 1967	Graphic Symbols for Electrical and Electronics Diagrams

(Application for copies should be addressed to DOD Single Stock Point, U.S. Naval Supply Depot, Philadelphia, Pennsylvania 19120.)

3. REQUIREMENTS

3.1 Description. The bandpass filter is a plug-in electronic assembly composed of various electronic components which are sealed in a metal container. Its purpose is to pass a narrow band of frequencies to the next stage, while rejecting or shunting to ground all other frequencies applied to its input terminals.

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3.2 General requirements

3.2.1 Data required. No data is required by this specification cation (other than ts accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the bandpass filter shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing bandpass filters that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the bandpass filter shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563728.

3.2.4 Surface treatment, construction, and dimensions. Surface treatment, construction, and dimensions for the bandpass filter shall be in accordance with the applicable drawings, specifications, and publications listed in NAVORD Dwg 1563728.

3.2.5 Moiture, fungus, and corrosion. The bandpass filter shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in NAVORD Dwg 1563728.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in NAVORD Dwg 1563728.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook M28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

303 Symbols. Interpret schematic diagram symbols in accordance with USAS Y32.2.

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3.4 Performance requirements and product characteristics. The bandpass filter shall meet the following performance requirements and product characteristics when tested in a test circuit as shown in Figure 1 or 2.

3.4.1 Input and output impedance. The input and output impedance of the bandpass filter shall be 200,000 plus or minus 10,000 ohms within the bandpass of the filter, when measured as specified in 4.4.4.11.

3.4.2 Direct current rating. The direct current rating of the input circuit (pins 1 and 7) shall be not less than 0.010 ampere, when measured as specified in 4.4.4.4.

3.4.3 Direct current resistance. The direct current resistance between pins 1 and 7 shall be less than 1000 ohms, when tested as specified in 4.4.4.5.

3.4.4 Bandwidth. The bandwidth of the filter shall be 2000 plus or minus 100 Hz, when measured as specified in 4.4.4.6.

3.4.5 Center frequency. The center frequency of the filter, when measured as specified in 4.4.4.7, shall be 7000 plus or minus 75 Hz.

3.4.6 Response. The response of the filter, when measured as specified in 4.4.4.8, shall be uniform within 1 decibel (db) from 6200 Hz to 7800 Hz.

3.4.7 Attenuation. The filter shall attenuate frequencies of 4 KHz and 13 KHz to a level which is 30 db or more below the level of the frequency of peak response, when measured as specified in 4.4.4.9.

3.4.8 Insertion loss. The insertion loss of the filter at center frequency shall be less than 1.5 db, when measured as specified in 4.4.4.10.

3.4.9 Insulation resistance. The insulation resistance of the filter shall be at least 10,000 megohms, when measured as specified in 4.4.4.3.

3.4.10 Dielectric strength. There shall be no evidence of breakdown between the following points, when tested as specified in 4.4.4.2: (1) any terminal (pin) and the case (mounting stud), arid (2) between either input terminal (pin 1 or pin 7) and either output terminal (pin 3 or pin 5).

3.4.11 Input voltage. The handpass filter shall meet the requirements of 3.3 with an input voltage of 100 vrms, 7000 Hz.

3.4.12 DC continuity. There shall be no dc continuity between input and output terminals of the bandpass filter.

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3.4.13 Operating life. The bandpass filter, when tested as specified in 4.5.5, shall have an operating life of at least 800 cycles of operation.

3.5 Environmental requirements. The bandpass filter shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.5.1 Vibration. The bandpass filter shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The bandpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.1

3.5.1.2 Vibration, operating. The bandpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.2.

3.5.1.3 Vibration, transportation. The bandpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.3.

3.5.2 Shock, transportation. The bandpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.5.3 Temperature and humidity

3.5.3.1 High temperature and humidity, nonoperating. The bandpass filter shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.5.3.2 High temperature, operating. The bandpass filter shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3.7.3.3 Low temperature, nonoperating. The bandpass filter shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

3.5.3.4 Low temperature, operating. The bandpass filter shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.4.

3.5.4 Immersion. The bandpass filter shall show no evidence of leakage during the immersion test, specified in 4.5.4, and thereafter shall meet the performance and product requirements of this specification.

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3.6 Cleanup. Prior to and after final assembly all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as excessive rosin flash, shall be removed.

3.7 Workmanship. The bandpass filter, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the bandpass filters shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

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4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five bandpass filters, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the bandpass filters by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five bandpass filters will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The bandpass filters shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each bandpass filter delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the bandpass filter shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18.33 degrees c to 35 degrees C).

4.4.1.2 Altitude. Normal ground.

4.4.1.3 Vibration. None.

4.4.1.4 Humidity. Room ambient to 95 percent relative, maximum.

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4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.4.3 Test equipment. The following test equipment is required for performing the quality conformance tests of the bandpass filter:

4.4.3.1 Oscillator: Western Electric 17B, or equivalent.

4.4.3.2 Electronic counter: Hewlett-Packard 521C, or equivalent.

4.4.3.3 Voltmeter: Ballantine Model 300, or equivalent.

4.4.3.4 Insulation tester: Radio Frequency Laboratories Model 184C or equivalent.

4.4.3.5 Dielectric tester: An instrument that will provide a 200 plus or minus 10 vdc test potential and a device to indicate a dielectric breakdown.

4.4.3.6 Resistors: 200,000 ohms plus or minus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B2003J; two required.

4.4.3.7 Power source: 115 volts, 60 HZ single phase.

4.4.3.8 power supply: Capable of delivering 90 vdc at a load current of 0.01 ampere with less than 3 percent ripple.

4.4.3.9 Resistor, variable: 10,000 ohms plus or minus 10 percent, 2 watts; MIL-R-94 designation RV2NAXD103A.

4.4,3.10 Millianmeter: O to 20 made, plus or minus 2 percent accuracy.

4.4.3.11 Resistor: 150,000 ohms plus or minus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B1503J.

4.4.3.12 Resistor: 180,000 ohms plus or minus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B1803J.

4.4.3.13 Resistor: 190,000 ohms plus or minus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B1903J.

4.4.3.14 Resistor: 210,000 ohms plus or minus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B2103HJ.

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4.4.3.15 Resistor: 220,000 chins plus or minus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B2203J.

4.4.3.16 Resistor: 250,000 ohms plus pr mimus 1 percent, 0.5 watt; MIL-R-10509 designation RN70B2503J.

4.4.3.17 Ohmmeter: Hewlett-Packard Model +1OB, or equivalent,

4.4.3.18 Connectors, test leads, and small tools: As required,

4.4.3.19 Voltmeter: Input impedance 11 megohms, shunted by 7.5 plus or minus 1 picofards; including test leads. Accuracy of reading within 3 percent.

4.4,4 Test procedure

4.4.4.1 Connect the two voltmeters (4.4.3.3 and 4.4.3.19), the oscillator (4.4.3.1), the electronic counter (4.4.3.2), the insulation tester (4.4.3.4), and the dielectric tester (4.4.3.5) to the power source (4.4.3.7), Set the power switches of these instruments to ON and allow them to warm up for the time necessary for them to stabilize.

4.4.4.2 To measure the dielectric withstanding voltage as set forth in 3.4.10:

4.4.4.2.1 Connect the dielectric tester (4.4.3.5) to pins 1 and 3 of the bandpass filter and apply 150 plus or minus 10 vdc for a period of one minute. At the end of the one minute period, there shall be no evidence of arcing, breakdown, or puncture.

4.4.4.2.2 Repeat 4.4.4.2.1 with the dielectric tester connected to pins 1 and 5. There shall be no evidence of arcing, breakdown, or puncture.

4.4.4.2.3 Repeat 4.4.4.2.1 with the dielectric tester connected to pin 1 and the case (mounting stud) of the bandpass filter. There shall be no evidence of arcing, breakdown, or puncture.

4.4.4.2.4 Repeat 4.4.4.2.3 for pins 2, 3, 4, 5, 6, and 7 of the band= pass filter. There shall be no evidence of arcing, puncture, or breakdown.

4.4.4.3 To measure the insulation resistance as set forth in 3.4.9:

4.4.4.3.1 Connect the insulation tester (4.4.3.4) to pins 1 and 3 of the bandpass filter and apply 100 plus or minus 10 vdc for a period of two minutes. At the end of the two minute period, measure the resistance at a test potential of 100 plus or minus 10 vdc. The resistance shall be at least 10,000 megohms.

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4.4.4.3.2 Repeat 4.4.4.3.1 with the insulation tester connected to pins 1 and 5, 7 and 3, and 7 and 5. The resistance in each case shall be at least 10,000 megohms.

4.4.4.3.3 Connect the insulation tester to pin 1 and the case (mounting stud) of the bandpass filter and apply 100 plus or minus 10 vdc for two minutes. At the end of the two minute period, measure the resistance at a test potential of 100 plus or minus 10 vdc. The resistance shall be at least 10,000 megohms.

4.4.4.3.4 Repeat 4.4.4.3.3 for pins 2, 3, 4, 5, 6, and 7 of the bandpass filter. In each case the resistance shall be at least 10,000 megohms.

4.4.4.4 To test the dc current rating as set forth in 3.4.2:

4.4.4.4.1 Connect the bandpass filter and the test equipment as shown in Figure 1. Adjust variable resistor R1 fully counterclockwise (maximum resistance). Set the power supply OFF/ON switch to ON and adjust the current for 10 milliamperes (ma), as indicated on milliammeter Ml.

4.4.4.4.2 Allow 10 made to flow for a period of 30 minutes.

4.4.4.4.3 After completion of 4.4.4.4.1 and 4.4.4.4.2, the bandpass filter shall meet the requirements of the remaining tests.

4.4.4.5 To test the dc resistance as set forth in 3.4.3:

4.4.4.5.1 Remove all connections from pins 1 and 7 of the bandpass filter.

4.4.4.5.2 Connect the ohmmeter (4.4.3.17) leads to pins 1 and 7 and set the ohmmeter range switch so that the meter needle is near half-scale. The resistance indicated shall be less than 1000 ohms.

4.4.4.6 To measure the bandwidth set forth in 3.4.4:

4.4.4.6.1 Connect the bandpass filter and test equipment (4.4.3) as shown in Figure 2, ensuring that the ground (low side) terminals of the test equipment are connected to pins 3 and 7 of the filter. To prevent stray pickup, careful placement of the leads is necessary.

4.4.4.6.2 Check the calibration of the electronic counter in accordance with the instructions provided with that instrument. When the counter has been satisfactorily checked, set the controls to measure frequency.

4.4.4.6.3 Set the oscillator frequency to 7 KHZ and adjust the output control for 1.0 vrms on the output voltmeter (the voltmeter connected between pins 5 and 3 of the filter).

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4.4.4.6.4 Adjust the oscillator frequency control until the output voltmeter indicates the maximum voltage. Record this frequency. Then readjust the oscillator output for 1.0 vrms, as indicated on the output voltmeter l

4.4.4.6.5 Record the indication on the input voltmeter (the voltmeter connected between pins 1 and 7 of the filter). Maintain this voltage at the filter input until otherwise instructed.

4.4.4.6.6 Decrease the oscillator frequency below 7 KHz until the output voltmeter indication is 3 db below 1 vrms. Record the frequency indicated on the electronic counter.

4.4.4.6.7 Increase the oscillator frequency above 7 KHz until the output voltmeter indication is again 3 db below 1 vrms. Record the frequency indicated on the electronic counter.

4.4.4.6.8 Subtract the frequency recorded in 4.4.4.6.6 from the frequency recorded in 4.4.4.6.7. The difference of these frequencies is the band width of the filter. The band width shall be 2000 plus or minus 100 Hz.

4.4.4.7 To measure the center frequency as set forth in 3.4.5.

4.4.4.7.1 Repeat the procedures of 4.4.4.6.1 through 4.4.4.6.7.

4.4.4.7.2 Add the frequencies recorded in 4.4.4.6.6 and 4.4.4.6.7 and divide the sum by two. This quotient is the center frequency of the bandpass filter. The center frequency shall be 7000 plus or minus 75 Hz.

4.4.4.8 To measure the response as set forth in 3.4.6.

4.4.4.8.1 Set the oscillator to the frequency recorded in 4.4.4.6.4. Adjust the oscillator output for 1 vrms on the output voltmeter.

4.4.4.8.2 Decrease the oscillator frequency below 7 KHz until the output voltmeter indicates 1 db below 1 vrms. Record the frequency at which this occurs. This frequency shall not be above 6200 Hz.

4.4.4.8.3 Increase the oscillator frequency above 7 Khz until an indication of 1 db below 1 vrms is again obtained on the output voltmeter. Record the frequency at which this occurs. This frequency shall not be below 7800 Hz.

4.4.4.9 To measure the attenuation as set forth in 3.4.7’:

4.4.4.9.1 Adjust the oscillator to the frequency recorded in 4.4.4.6.4. Set the oscillator output to 1 vrms, as indicated on the output voltmeter.

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4.4.4.9.2 Decrease the oscillator frequency below 7 KHz until the output voltmeter indicates 30 db below 1 vrms. When this indication is obtained, record the frequency indicated on the electronic counter. This frequency shall not be lower than 4 KHz.

4.4.4.9.3 Increase the oscillator frequency above 7 KHz until the output voltmeter again indicates 30 db below 1 vrms and record the frequency indicated on the electronic counter. This frequency shall not be greater than 13 KHz.

4.4.4.10 To measure the insertion loss as set forth in 3.4.8:

4.4.4.10.1 Adjust the oscillator frequency control until the electronic counter indicates the center frequency determined in 4.4.4.7.2. Move the input voltmeter probe to pin 1 of the filter. Then set the oscillator output for 1 vrms as indicated on the output voltmeter.

4.4.4.10.2 Record the voltage indicated on the input voltmeter. Compute the insertion loss in decibels from the following expression:

Insertion loss (in db) = $20 \log_{10} V$

where V is the voltage recorded in 4.4.4.10.2.

The insertion loss shall be less than 1.5 db.

4.4.4.11 To measure the input and output impedance as set forth in 3.4.1:

4.4.4.11.1 Set the oscillator frequency to the center frequency determined in 4.4.4.7.2. Set the oscillator output to 1 vrms, as indicated on the output voltmeter. Using the expression:

$$\text{Power} = \frac{V^2}{R} \text{ where}$$

V = voltage indicated on output voltmeter, and
R = load resistance,

compute the power dissipated in the load resistance.

4.4.4.11.2 Without adjusting the input voltage, change the load resistance to 0.150 megohm (4.4.3.11), 0.180 megohm (4.4.3.12), 0.190 megohm (4.4.3.13), 0.210 megohm (4.4.3.14), 0.220 megohm (4.4.3.15), and 0.250 megohm (4.4.3.16). For each resistance, compute the power dissipated in the load resistance, using the expression given in 4.4.4.11.1. Maximum

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power dissipation (indicating R_{Load} = output impedance) shall occur using a load resistance of 200,000 plus or minus 1 .000 ohms.

-. .-.11.3 Reverse the input and output connections to the bandpass filter (i.e., use terminals 3 and 7 for the input and terminals 1 and 7 for the output). Terminals 3 and 7 shall remain grounded.

4.4.4.11.4 . 4.4. 4. 11.2 with the filter connected as specified in 4.4.4.11.3. Again, maximum power dis sipation shall occur using a load re si stance of 200,000 plus or minus 10,000 ohms.

4. 4. 4. 12 Visual and dimensional examination. Each bandpass filter shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.13 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Vibration, nonoperating. The bandpass filter, mounted as specified in Figure 3, shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

4.5.1.2 Vibration, operating. The bandpass filter, mounted as specified in Figure 3, shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.5.1.2.

4-.5.1.3 Vibration, transportation. The bandpass filter shall be subjetted to the Vibration, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.3.

4.5.2 Shock, transportation. The bandpass filter shall be subjected to the "Shock, transportation test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.2.

4.5.3 Temperature and humidity

4.5.3.1 High temperature and humidity, nonoperating. The bandpass filter shall be subjected to the "High temperature and humidity, non-

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operating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.1.

4.5.3.2 High temperature, operating. The bandpass filter shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.2.

4-.5.3.3 Low temperature, nonoperating. The bandpass filter shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.3.3.

4.5.3.4 Low temperature, operating. The bandpass filter shall be subjetted to the "Low temperature, operating" test of' MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.5.3.4.

4.5.4 Immersion. The bandpass filter shall show no evidence of leakage while being subjected to the immersion test specified in MIL-STD-202, Method 104, Condition A, and thereafter shall meet the performance and product requirements of this specification.

4.5.5 Operating life. The test for operating life shall be conducted as follows:

4.5,5.1 Connect the oscillator subassembly as shown in Figure 1. Adjust the variable resistor for 10 ma of current and allow this current to flow for 30 plus or minus 1 minutes. Disconnect the 90 vdc and allow the circuit to rest for at least 1 hour. This completes one cycle of operation.

4.5.5.2 Repeat 4.5.5.1 until 800 cycles of operation have been completed, after which the bandpass filter shall meet all the requirements of 3.4.1 through 3.4.12.

4.5.5.3 The test circuit components maybe replaced as necessary to allow continuance of the test.

4.5.6 Functional tests. At the conclusion of the tests specified in 4.5.1.3 and 4.5.2 above, functional and other tests, as applicable, shall be conducted on the bandpass filter to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics of this specification.

4.5.7 Waiver of tests. Tests specified in 4.5.1.3 and 4.5.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be specified (see 6.2).

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5.1.1 Level A

5.1.1.1 Unit packaging. The bandpass filter shall be packaged in accordance with Method IIB of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The bandpass filter shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The bandpass filter, packaged as described in 5.1.1.1, shall be packed in a wood box in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 101 (MS 53102) (see 6.2).

5.2.2 Level C. The bandpass filter, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Level A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the bandpass filter will meet all the requirements of Section 3 after being tested in accordance with 4.5.1.3 (Vibration) and 4.5.2 (shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than five percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

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6. NOTES

6.1 Intended use. The bandpass filter is intended for use in the Torpedo Mark 37 Mods 0, 2, and 30

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) Guarantee clause (see 6.4).

(h) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to 4.5.7.

6.3 preparation for delivery criteria. Criteria for use of the proper level of preservation packaging. and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. The contract or order shall include a requirement for a guarantee stating that any bandpass filter which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure

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to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

Preparing Activity
Navy - CS
Project No. 5915-N002

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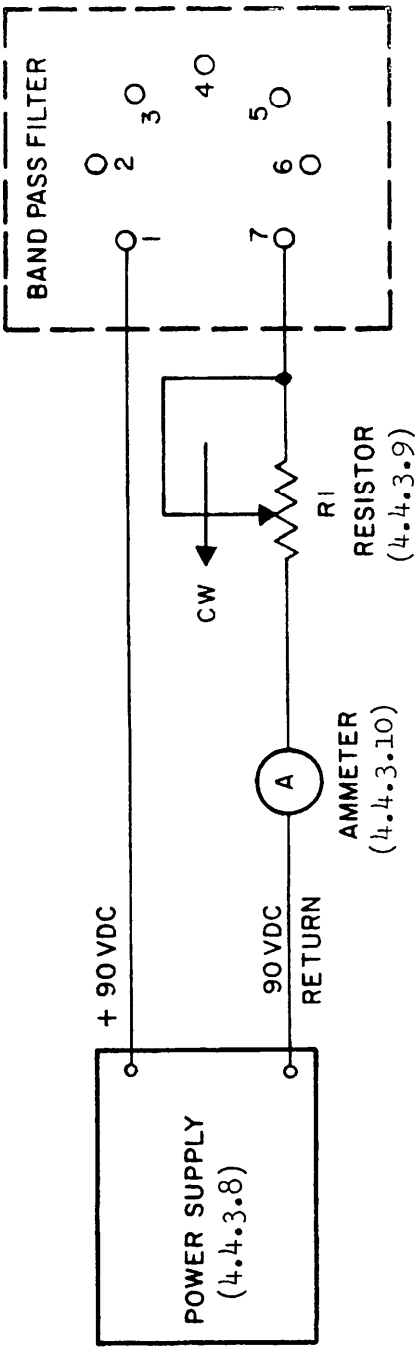
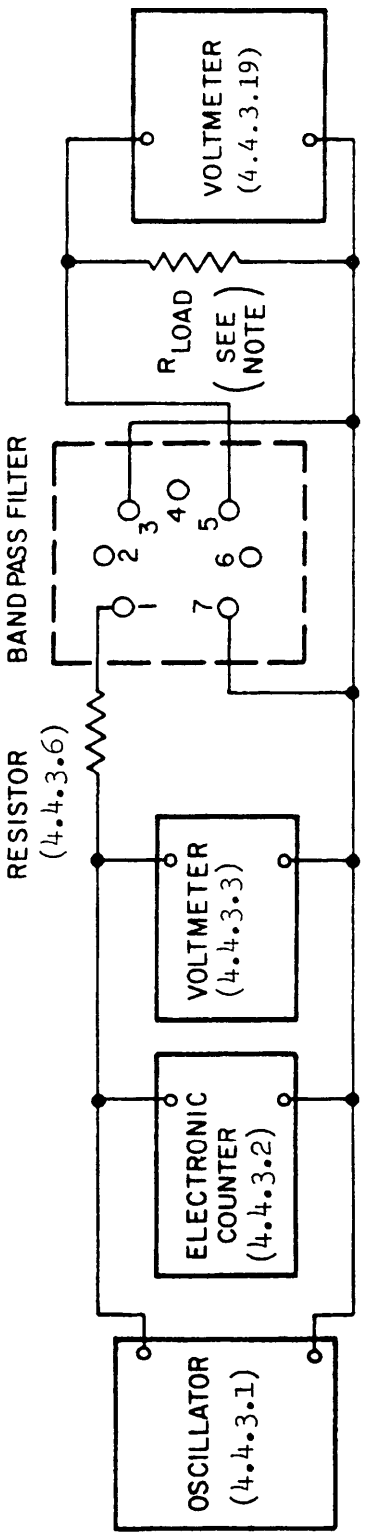


Figure 1. Bandpass Filter DC Current Rating Test, Schematic Diagram.

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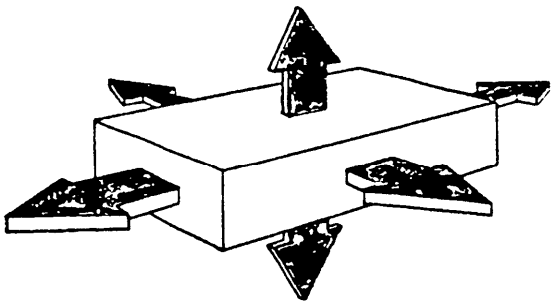


NOTE: R_{LOAD} is 200,000 ohms ±1 percent (4.4.3.6) except in 4.4.4.11.2, where the ±1 percent resistors (4.4.3.11 through 4.4.3.15) of different values are substituted.

Figure 2. Bandpass Filter Test Circuit, Schematic Diagram.

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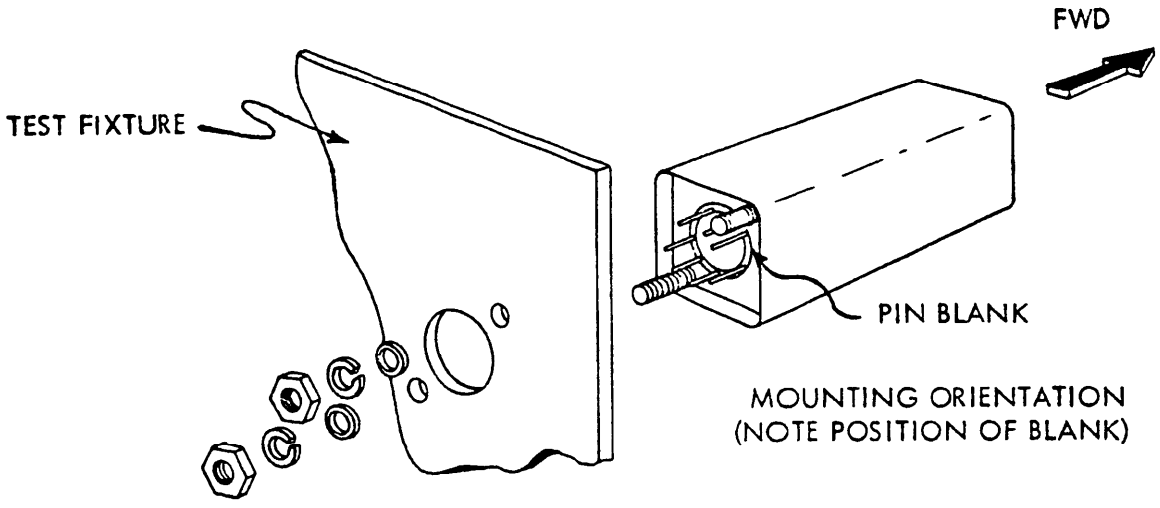
FIGURE 3. FILTER NAVORD DWG 15637[illegible]8
MOUNTING REQUIREMENTS



EARTH'S GRAVITATIONAL FIELD



DIRECTIONS OF SHOCK AND VIBRATION INPUTS



NOTES:

1. FOR VIBRATION, ORIENTATION OF TEST ITEM WITH RESPECT TO EARTH'S GRAVITATIONAL FIELD SHALL BE MAINTAINED FOR ALL TEST DIRECTIONS.
2. FOR SHOCK, ORIENTATION OF THE TEST ITEM CAN BE CHANGED AS NEEDED TO ACCOMPLISH ALL DIRECTIONS OF TEST.
3. FOR BOTH SHOCK AND VIBRATION, THE TEST ITEM SHALL BE SECURED TO THE TEST FIXTURE AS SHOWN.
4. THE TEST FIXTURE SHALL NOT ALTER THE INPUTS FROM SHOCK OR VIBRATION EXCITER TO THE MOUNTING POINTS OF THE TEST ITEM.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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